

Dylan Smith Homework # 10

2.	SS	df	MS	F
Treatment	93.33	2	46.67	8.75
Error	64.00	12	5.33	
Total	157.33	14		

3. $H_0: \mu_1 = \mu_2 = \mu_3 = 0$ $\alpha = 0.03$

H_1 : At least one $\mu_i \neq 0$

p-value = 0.004530596

F = 8.75

Decision: Reject H_0 . $p = 0.004530596 < 0.03 = \alpha$

Conclusion: There is sufficient evidence to conclude that not all treatments are equal.

4. $\bar{y}_{10} = 12$ $\bar{y}_{50} = 14$ $12 - 14 = -2$

MSE = 5.33 $n = 5$

SE = $\sqrt{5.33 \left(\frac{1}{5} + \frac{1}{5} \right)} = 1.46$ $\text{invT}(0.015, 12) = 2.46$

$-2 \pm 2.46(1.46)$

CI = $(-5.59, 1.59)$

We cannot conclude that the means are different because 0 is inside the interval.

$$5. \quad \bar{y}_{10} = 12 \quad \bar{y}_{25} = 18 \quad \bar{y}_{50} = 14$$

$$MSE = 5.33 \quad df_{error} = 12 \quad n = 5$$

$$q \text{ Tukey } (0.97, 3, 12) = 4.18$$

$$4.18 \cdot \sqrt{\frac{5.33}{5}} \cdot \frac{1}{\sqrt{2}} = 3.05$$

$$\mu_{10} - \mu_{25} = 12 - 18 = -6$$

$$-6 \pm 3.05$$

$$(-9.05, -2.95)$$

Significant because CI does not contain 0.

$$\mu_{10} - \mu_{50} = 12 - 14 = -2$$

$$-2 \pm 3.05$$

$$(-5.05, 1.05)$$

Not significant because CI does contain 0.

$$\mu_{25} - \mu_{50} = 18 - 14 = 4$$

$$4 \pm 3.05$$

$$(0.95, 7.05)$$

Significant because CI does not contain 0.

$$b. \sum_{i=1}^k n_i (\bar{y}_i - \bar{\bar{y}})^2 = \sum_{i=1}^k n_i (\bar{y}_i^2 - 2\bar{y}_i \bar{\bar{y}} + \bar{\bar{y}}^2)$$

$$= \sum_{i=1}^k n_i \bar{y}_i^2 - 2\bar{\bar{y}}(N\bar{\bar{y}}) + N\bar{\bar{y}}^2$$

$$= \sum_{i=1}^k n_i \bar{y}_i^2 - N\bar{\bar{y}}^2$$

$$N\bar{\bar{y}}^2 = N \left(\frac{1}{N} \sum_{i=1}^k \sum_{j=1}^{n_i} y_{ij} \right)^2 = \frac{(\sum_{i=1}^k \sum_{j=1}^{n_i} y_{ij})^2}{N}$$

$$\sum_{i=1}^k n_i (\bar{y}_i - \bar{\bar{y}})^2 = \sum_{i=1}^k \frac{(\sum_{j=1}^{n_i} y_{ij})^2}{n_i} - \frac{(\sum_{i=1}^k \sum_{j=1}^{n_i} y_{ij})^2}{N}$$